SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology | Department of Agricultural Engineering

AEN 214: INTRODUCTION TO ELECTRONICS — LECTURE 8

Introduction to Digital Electronics

Logic Gates, Universal Gate Synthesis, Propagation Delay, Adders, Subtractors & Latches

Course Instructor: Dr. Kadeghe Goodluck Fue, PhD, P.Eng (T)

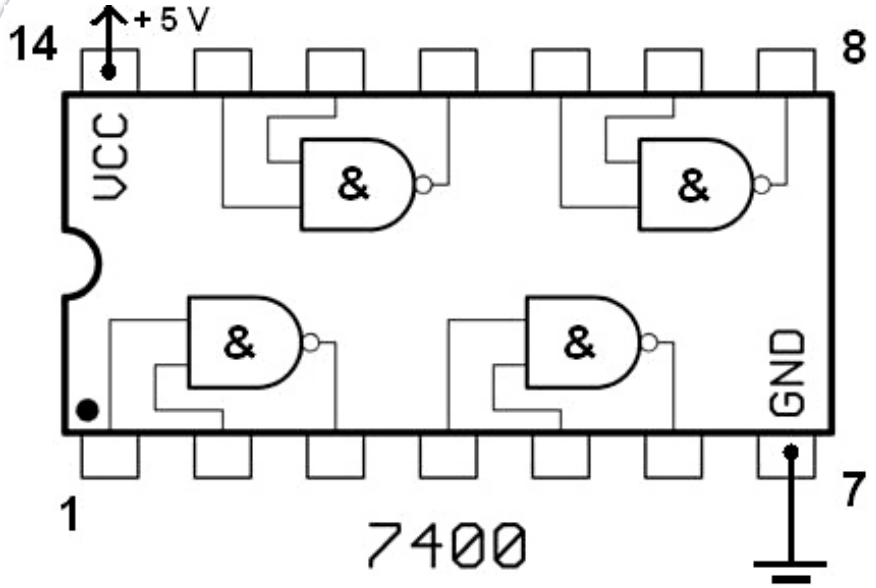
Electronics & Precision Agriculture Laboratory | Academic Year 2025/2026 | Sokoine University of Agriculture

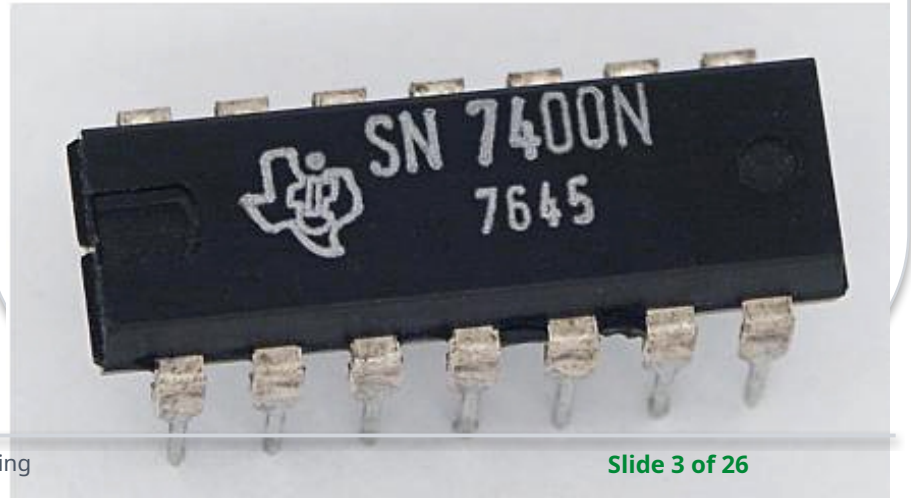
Lecture 8 Learning Outcomes & Combinational Scope

- **Master Primary Logic Gates:** Analyze truth tables, Boolean expressions, and IEEE symbols for AND, OR, and NOT gates.
- **Evaluate Secondary & Universal Gates:** NAND, NOR, XOR, and XNOR gate operations and parity testing.
- **Synthesize Universal Gate Circuits:** Implement NOT, AND, OR, NOR, and XOR using purely NAND or purely NOR gates.
- **Analyze Digital Electrical Parameters:** Propagation delay t_pd, rise/fall times, noise margins, and power dissipation.
- **Design Combinational Arithmetic Circuits:** Half Adder, Full Adder, Ripple Carry Adder, and Half/Full Subtractors.
- **Implement Multi-Condition Agricultural Interlocks:** Greenhouse climate control, grain elevator safety, and pump priming.
- **Introduce Sequential Logic Elements:** Bistable multivibrators, active-LOW SR latches, and switch debouncers.

Digital Logic in Agricultural Machinery Automation

- **The Decision-Making Core:** Logic gates form the hardware decision engine of every farm machinery safety and control system.
- **Fail-Safe Hardware Interlocks:** Hardwired logic guarantees emergency shutdown of grain augers even if software crashes.
- **Real-Time Response:** Discrete hardware logic gates respond in nanoseconds, preventing mechanical operator injury.
- **Universal Gate Flexibility:** Remote farm workshops can replace any damaged logic chip using abundant 74LS00 Quad NAND ICs.
- **Combinational Sensor Processing:** Combines inputs from soil moisture, ambient humidity, and wind speed into pump commands.



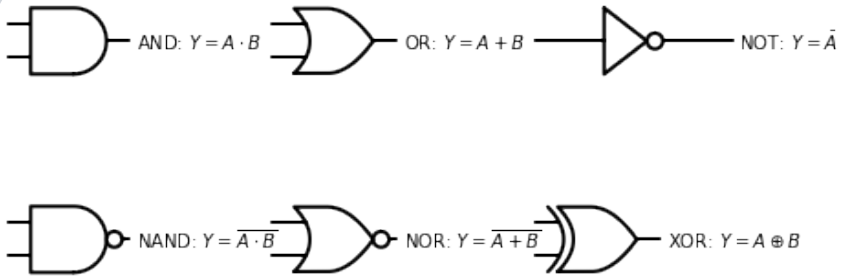


Definition of a Logic Gate & Voltage Thresholds

- **Definition:** An electronic circuit with one or more binary inputs and a single binary output that executes a Boolean logical function.
- **Binary Voltage Abstraction:** Digital circuits map continuous electrical voltages into discrete binary logic states: Logic 0 (LOW) and Logic 1 (HIGH).
- **TTL Voltage Standards (5V Supply):** $V_{IL(max)} = 0.8$ V, $V_{IH(min)} = 2.0$ V, $V_{OL(max)} = 0.4$ V, $V_{OH(min)} = 2.4$ V.
- **CMOS Voltage Standards (5V Supply):** $V_{IL(max)} = 1.5$ V, $V_{IH(min)} = 3.5$ V, $V_{OL(max)} = 0.05$ V, $V_{OH(min)} = 4.95$ V.
- **Indeterminate Zone:** Voltages between V_{IL} and V_{IH} are invalid; gates may oscillate or draw excessive current in this region.

Primary Logic Gates: AND, OR, and Inverter (NOT)

- **AND Gate:** Output is HIGH if and only if ALL inputs are HIGH: Y = A · B. Equivalent to series electrical contacts.
- **OR Gate:** Output is HIGH if ANY input is HIGH: Y = A + B. Equivalent to parallel electrical contacts.
- **NOT Gate (Inverter):** Output is the exact opposite of the input: Y = /A. Single-input logic negation.
- **Truth Tables:** Mathematical matrices listing all 2^N possible binary input combinations and resulting output states.
- **IEEE Distinctive Shape Symbols:** AND has flat input face with curved output; OR has curved input face with pointed output; NOT is triangle with bubble.



Secondary Logic Gates: NAND, NOR, XOR, and XNOR

- **NAND Gate (NOT-AND):** AND followed by an inverter: Y = /(A · B). Output is LOW only when ALL inputs are HIGH.
- **NOR Gate (NOT-OR):** OR followed by an inverter: Y = /(A + B). Output is HIGH only when ALL inputs are LOW.
- **XOR Gate (Exclusive-OR):** Output is HIGH when inputs are DIFFERENT: Y = A ⊕ B = /A·B + A·/B. Senses parity and inequalities.
- **XNOR Gate (Exclusive-NOR):** Output is HIGH when inputs are IDENTICAL: Y = /(A ⊕ B) = A·B + /A·/B. Digital magnitude comparator.
- **Active-LOW Bubbles:** Inversion bubbles at inputs or outputs denote active-LOW signal polarity.

The Universal NAND Gate: Complete Functional Completeness

- **Functional Completeness:** Any Boolean logic expression can be implemented using NOTHING but 2-input NAND gates!
- **Synthesizing an INVERTER:** Tie both inputs of a NAND gate together: Y = /(A · A) = /A.
- **Synthesizing an AND Gate:** Follow a NAND gate with an Inverter: Y = / (/(A · B)) = A · B (Requires 2 NAND gates).
- **Synthesizing an OR Gate:** Invert inputs A and B before feeding into NAND: Y = /(/A · /B) = //A + //B = A + B (via De Morgan; 3 NAND gates).
- **Synthesizing a NOR Gate:** Follow the synthesized OR gate with an inverter: Y = /(A + B) (Requires 4 NAND gates).
- **Agricultural Supply Chain Impact:** Technicians need to stock only 74LS00 or CD4011 Quad 2-input NAND ICs to repair any farm logic board!

The Universal NOR Gate: Functional Completeness

- **Universal NOR Architecture:** Dual to the NAND gate; any Boolean expression can be realized using NOR gates alone.
- **Synthesizing an INVERTER:** Tie both inputs of a NOR gate together: Y = /(A + A) = /A.
- **Synthesizing an OR Gate:** Follow a NOR gate with a NOR inverter: Y = / (/(A + B)) = A + B (Requires 2 NOR gates).
- **Synthesizing an AND Gate:** Invert inputs before feeding into NOR: Y = /(/A + /B) = //A · //B = A · B (via De Morgan; 3 NOR gates).
- **Synthesizing a NAND Gate:** Follow the synthesized AND gate with an inverter: Y = /(A · B) (Requires 4 NOR gates).
- **Silicon Fab Preference:** In CMOS technology, NAND gates are faster than NOR gates due to high electron mobility in N-channel pull-down networks.

Digital Dynamic Parameters & Noise Margins

- **Propagation Delay (t_pd):** Finite time elapsed between input transition and output response; t_pd = 0.5 · (t_pLH + t_pHL) (typically 5 to 30 ns).
- **Rise Time (t_r) & Fall Time (t_f):** Time required for output voltage to transition between 10% and 90% of steady-state logic levels.
- **High-State Noise Margin:** NM_H = V_OH(min) - V_IH(min). For standard TTL: NM_H = 2.4V - 2.0V = 0.40 V.
- **Low-State Noise Margin:** NM_L = V_IL(max) - V_OL(max). For standard TTL: NM_L = 0.8V - 0.4V = 0.40 V.
- **CMOS Superiority:** CMOS noise margins approach 30% to 40% of supply voltage (NM ≈ 1.5V at 5V V_DD); vastly superior in noisy farm environments!

Worked Engineering Example 1: Universal NAND Realization of Dairy Milk Vat Interlock

PROBLEM SPECIFICATION & AGRICULTURAL CONTEXT:
A dairy milk chilling tank agitator motor in Arusha operates according to Boolean logic: Y = A · B + /C, where A = Milk Level Normal, B = Temperature Above 4°C, and C = Tank Wash Cycle Active. To simplify spares inventory, the farm workshop manager insists on building the entire circuit using standard 74HC00 Quad 2-Input NAND ICs. (a) Express Y using only NAND operators via De Morgan's theorem, (b) Draw the NAND gate interconnection, (c) Determine the minimum number of 74HC00 ICs required.

STEP-BY-STEP MATHEMATICAL SOLUTION & VERIFICATION:

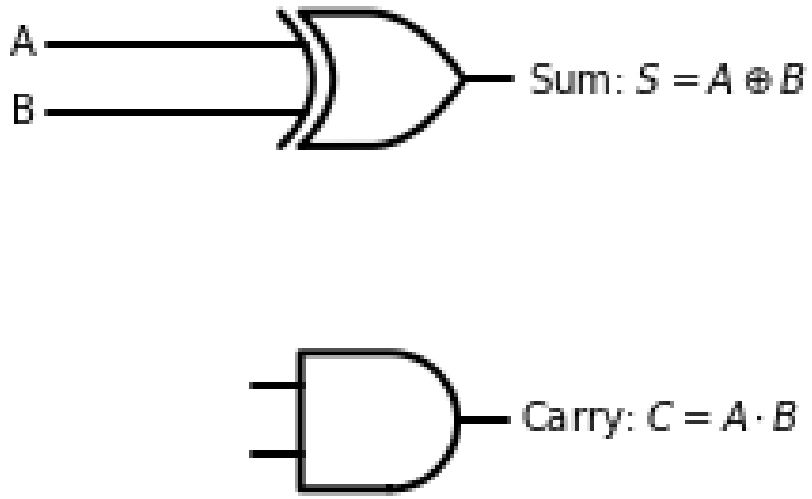
1. Apply Double Negation (Involution): Y = //(A · B + /C).
2. Apply De Morgan's Law to Inner Expression: / ((A · B) + /C) = /(A · B) · //C = /(A · B) · C.
3. Complete Inverted Expression: Y = / [/(A · B) · C].
4. Verify NAND Realization:
   - Gate 1 (2-input NAND): Inputs A and B -> Output X_1 = /(A · B).
   - Gate 2 (2-input NAND): Inputs X_1 and C -> Output Y = / [X_1 · C] = / [/(A · B) · C] = A · B + /C.
5. Circuit Gate Count & Package Requirement:
   - Total 2-input NAND gates required = EXACTLY TWO GATES!
   - A single 74HC00 IC contains FOUR independent 2-input NAND gates.
   - Exactly ONE 74HC00 IC is required, with two spare gates available for future automation expansion.

Combinational Arithmetic: The Half Adder

- **Circuit Function:** Adds two single-bit binary inputs (A and B) and produces a Sum bit (S) and an outgoing Carry bit (C).
- **Truth Table Analysis:** S is HIGH when inputs differ (0+1=1, 1+0=1); C is HIGH only when both inputs are 1 (1+1=0, Carry 1).
- **Boolean Formulations:** Sum: $S = A \oplus B = /A \cdot B + A \cdot /B$; Carry: $C = A \cdot B$.
- **Hardware Implementation:** Synthesized using exactly one 2-input XOR gate (for Sum) and one 2-input AND gate (for Carry).
- **Half Adder Limitation:** Lacks an incoming carry input (C_in); cannot be cascaded to add multi-bit binary numbers.

Combinational Arithmetic: The Full Adder

- **Circuit Function:** Adds three single-bit inputs: operands A and B, plus incoming carry C_in from previous lower stage.
- **Full Adder Boolean Equations:** Sum: S = A ⊕ B ⊕ C_in; Outgoing Carry: C_out = A · B + C_in · (A ⊕ B).
- **Two-Half-Adder Architecture:** Full adder is synthesized by cascading two Half Adders and an OR gate.
- **Half Adder 1:** S_1 = A ⊕ B; C_1 = A · B.
- **Half Adder 2:** S = S_1 ⊕ C_in = (A ⊕ B) ⊕ C_in; C_2 = S_1 · C_in = (A ⊕ B) · C_in.
- **Final Carry:** C_out = C_1 + C_2 = A · B + (A ⊕ B) · C_in.
- **Transistor Count:** CMOS transmission gate full adder requires only 28 transistors.



Parallel Multi-Bit Ripple-Carry Adder (74LS283)

- **N-Bit Parallel Adder:** Cascades N Full Adder stages in series to add two N-bit binary words simultaneously.
- **Carry Propagation Path:** Outgoing carry C_out of stage k connects directly to incoming carry C_in of stage k+1.
- **The Ripple-Carry Bottleneck:** Most Significant Bit sum (S_N-1) cannot be computed until the carry ripples through all preceding N stages!
- **Total Addition Delay:** $t_{total} = (N - 1) \cdot t_{carry} + \max(t_{carry}, t_{sum})$; For 32 bits at 10 ns/stage, delay reaches 320 ns.
- **Carry Lookahead Adder (CLA):** Fast adder that pre-computes carries in parallel using Generate ($G = A \cdot B$) and Propagate ($P = A+B$) logic.

Binary Subtractors: Half & Full Subtractor Circuits

- **Half Subtractor:** Subtracts single-bit subtrahend B from minuend A, producing Difference (D) and Borrow-out (B_out).
- **Half Subtractor Equations:** Difference: D = A ⊕ B; Borrow-out: B_out = /A · B.
- **Full Subtractor:** Computes A - B - B_in, taking into account incoming borrow B_in from lower significant bit.
- **Full Subtractor Equations:** Difference: D = A ⊕ B ⊕ B_in; Borrow-out: B_out = /A · B + B_in · /(A ⊕ B).
- **Adder-Subtractor Combined Circuit:** Place XOR gates on input B lines controlled by mode signal M (M=0 -> Add; M=1 -> 2's Complement Subtract!).

Worked Engineering Example 2: Propagation Delay Analysis in Tractor Autopilot Adder

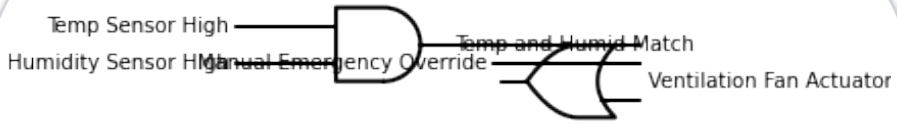
PROBLEM SPECIFICATION & AGRICULTURAL CONTEXT:
An autonomous tractor GPS navigation controller utilizes an 8-bit parallel ripple-carry adder (cascaded 74HC283 full adders) to compute real-time coordinate offsets. Datasheet timing specifications: Carry-in to Carry-out propagation delay t_carry = 12.0 ns; Carry-in to Sum bit delay t_sum = 16.0 ns; Operand input (A, B) to Carry-out delay t_gate = 18.0 ns. Determine: (a) Total time required to generate the final MSB carry C_8, (b) Total time required for all 8 sum bits to settle into valid steady state, (c) Maximum operating clock frequency of the ALU.

STEP-BY-STEP MATHEMATICAL SOLUTION & VERIFICATION:

1. Time to Generate Final Carry C_8: Stage 0 generates C_1 from operands A_0, B_0 in t_gate. Subsequent 7 stages ripple carry through t_carry.
   t_C8 = t_gate + (8 - 1) · t_carry = 18.0 ns + (7 · 12.0 ns) = 18.0 ns + 84.0 ns = 102.0 ns.
2. Time for All Sum Bits to Settle Validly:
   • Sum bits S_0 through S_6 settle earlier.
   • Final sum bit S_7 settles after carry arrives at stage 7 plus the carry-to-sum delay: t_settle = t_gate + (8 - 2) · t_carry + t_sum.
   t_settle = 18.0 ns + (6 · 12.0 ns) + 16.0 ns = 18.0 + 72.0 + 16.0 = 106.0 ns.
3. Worst-Case Computational Latency: Total settling latency t_max = max(t_C8, t_settle) = 106.0 ns.
4. Maximum ALU Clock Frequency: f_max = 1 / t_max = 1 / (106.0 x 10^-9 s) = 9.434 x 10^6 Hz = 9.43 MHz.
5. Engineering Assessment: To achieve 50 MHz autopilot processing, the design must upgrade from ripple-carry to a Carry-Lookahead Adder (CLA).

Multi-Sensor Combinational Control in Greenhouses

- **Autonomous Greenhouse Climate Controller:** Regulates evaporative cooling pad pumps, exhaust fans, and motorized ridge vents.
- **Sensor Inputs:** Temperature High (T: > 32°C), Humidity Low (H: < 50%), Solar Radiation High (S: > 800 W/m^2), Wind High (W: > 40 km/h).
- **Control Output:** Evaporative Cooling Pump (P): Activate when temperature is high, humidity is low, AND wind is not dangerous.
- **Boolean Function Synthesis:** P = T · /H · /W; Ridge vents must close automatically during severe winds: V_close = W + /T · /S.
- **Hardware Realization:** Hardwired logic gate networks execute safety shutdowns without depending on microcontroller firmware.



Worked Engineering Example 3: Fail-Safe Combinational Logic for Grain Silo Conveyor

PROBLEM SPECIFICATION & AGRICULTURAL CONTEXT:
Design a fail-safe digital interlock circuit for a 50-tonne/hour vertical bucket elevator at the Morogoro grain storage depot. The elevator drive motor must START (Y = 1) only if ALL of the following safety conditions are satisfied: (1) Main emergency stop pushbutton is ARMED (E = 1), (2) Discharge chute is CLEAR of grain blockage (C = 1), and (3) EITHER the Manual Start button is pressed (M = 1) OR the Automated Bin Level Controller requests fill (A = 1). In addition, if the Belt Slip sensor detects belt slippage (S = 1), the motor must IMMEDIATELY TRIP (Y = 0) regardless of all other inputs. (a) Formulate the Boolean expression, (b) Synthesize using standard logic gates.

STEP-BY-STEP MATHEMATICAL SOLUTION & VERIFICATION:

1. Analyze Safety Conditions & Assign Variables: Motor runs (Y = 1) if: E = 1, C = 1, (M = 1 OR A = 1), and S = 0 (Slip absent).
2. Formulate Boolean Logic Expression: $Y = E \cdot C \cdot (M + A) \cdot /S$.
3. Gate Synthesis Strategy:
   - Stage 1 (OR gate): Inputs M and A -> Output $X_1 = (M + A)$.
   - Stage 2 (Inverter): Input S -> Output $X_2 = /S$.
   - Stage 3 (4-input AND gate): Inputs E, C, X_1, X_2 -> Output $Y = E \cdot C \cdot (M + A) \cdot /S$.
4. Alternative All-NAND Synthesis: Apply De Morgan: $Y = / [/(E \cdot C) + /X_1 + S]$.
5. Fail-Safe Wiring: Emergency Stop (E) is wired normally-closed (NC) so that a severed wire automatically trips E = 0, halting the conveyor!

Introduction to Sequential Logic: The Bistable SR Latch

- **Combinational vs Sequential:** Combinational outputs depend strictly on present inputs; Sequential outputs depend on present inputs AND past history (memory).
- **Active-LOW NAND SR Latch:** Cross-coupled 2-input NAND gates with active-LOW inputs /Set (/S) and /Reset (/R).
- **Operating States:** /S=0, /R=1 -> Set (Q=1); /S=1, /R=0 -> Reset (Q=0); /S=1, /R=1 -> Hold / Memory (Q remains unchanged).
- **Forbidden (Invalid) State:** /S=0, /R=0 forces both Q=1 and /Q=1; violating complementary output definition; causes race when released.
- **Mechanical Switch Debouncing:** Mechanical pushbuttons bounce for 5 to 20 ms, generating dozens of false digital clock pulses.
- **SR Debouncer Action:** An SPDT switch with an SR latch latches cleanly on first contact bounce, providing a single pristine digital pulse.

Hardening Digital Logic in Harsh Farm Environments

- **Unused Input Pin Rule:** NEVER leave CMOS logic input pins floating (acts as high-impedance antenna, picking up EMI and causing gate latchup).
- **Pull-up and Pull-down Resistors:** Terminate all unused CMOS inputs solidly to V_DD or GND via 10 kOhm resistors.
- **Power Rail Decoupling:** Place a 0.1 microFarad ceramic capacitor directly adjacent to the V_CC pin of every single logic IC on the PCB.
- **Inductive Suppression:** Farm relays driven by logic outputs must have flyback diodes and optical isolators to block 500V kickbacks.
- **Schmitt-Trigger Inputs:** Specify 74HC14 or 74HCT14 Schmitt-trigger inverters on all external sensor lines to clean up slow, noisy edge transitions.

Tanzanian Field Case Study 1: Switch Debouncing Failure in Dairy Milking Pulsator

FIELD LOCATION & PROBLEM IDENTIFICATION

- **Geographic Location: Msolwa Dairy Farm Automated Milking Parlor, Kilombero, Tanzania**

Automated dairy milking units featured digital pulse counters to measure individual cow milk yield via a tipping-bucket mechanism (50 ml per tip). The digital counters registered over 300% false milk yield, reading 45 liters per cow instead of 12 liters and corrupting dairy management breeding records.

SUA ENGINEERING SOLUTION & AGRONOMIC IMPACT

SUA engineers hooked a digital oscilloscope to the mechanical magnetic reed switch on the tipping bucket. Oscilloscope traces revealed severe contact bounce: each physical bucket tip generated 8 to 14 electrical bounces spanning 12 milliseconds. The high-speed digital counter registered every single bounce as an independent 50 ml delivery. Engineers installed an active-LOW cross-coupled NAND SR latch (74HC00) debouncer.

Measurable Performance Metrics:

✔️ Completely eliminated contact bounce; exactly ONE pulse registered per mechanical tip.

✔️ Milk measurement accuracy restored to within ±0.5% of verified weight.

✔️ Saved the cooperative over 16,000 USD in misallocated nutritional feed supplements.

Tanzanian Field Case Study 2: Fail-Safe Logic Interlock for Grain Silo Deflagration

FIELD LOCATION & PROBLEM IDENTIFICATION

• Geographic Location: National Food Reserve Agency (NFRA) Complex, Morogoro, Tanzania

Grain elevators handle dry maize that generates explosive organic dust clouds. A bucket elevator leg bearing overheated to 180°C, igniting suspended corn dust and blowing off explosion relief panels. The PLC software loop lagged by 1.2 seconds, failing to shut off the feed auger in time.

SUA ENGINEERING SOLUTION & AGRONOMIC IMPACT

Engineers bypassed the software PLC for critical life-safety functions, installing a dedicated hardwired combinational logic tripping network using MIL-spec 74LS series gates. The circuit inputs: Bearing Temperature Sensor (B), Belt Alignment Sensor (A), Motor Current Overload (I), and Fire Suppression Pressure (P). The hardwired circuit trips the 100A main vacuum contactor within 18 nanoseconds of limit exceedance.

Measurable Performance Metrics:

✔️ Emergency tripping response time accelerated from 1.2 seconds down to 18 nanoseconds.

✔️ Successfully halted feed auger during a subsequent bearing failure, preventing an explosion.

✔️ Earned full safety compliance certification from OSHA Tanzania.

Laboratory Bench Protocol: Digital Logic Gates & Universal Synthesis Bench Protocol

PRACTICAL LAB LEARNING OUTCOMES & BENCH REQUIREMENTS:

- Verify truth tables of 74LS08 (AND), 74LS32 (OR), 74LS04 (NOT), and 74LS00 (NAND) ICs on breadboard.
- Synthesize an OR gate and an XOR gate using purely 74LS00 Quad 2-Input NAND gates.
- Assemble a Half Adder circuit using 74LS86 (XOR) and 74LS08 (AND) and verify Sum and Carry truth table.
- Build a switch debouncer using cross-coupled NAND gates and observe bounce suppression on DSO.

STEP-BY-STEP PROTOTYPING & DIAGNOSTIC PROCEDURE:

Step 1: Insert 74LS00, 74LS08, 74LS86 ICs onto breadboard. Connect Pin 14 to +5.0V and Pin 7 to GND. Add 0.1 uF decoupling caps.

Step 2: Connect input switches to logic 0 (GND) and logic 1 (+5V). Connect gate outputs to red LEDs with 330 Ohm resistors.

Step 3: Sweep all 4 input combinations; record output LED states; verify manufacturer truth tables.

Step 4: Wire the Universal NAND circuit for Y = A + B (3 NAND gates); prove that output matches standard OR gate.

Step 5: Connect noisy mechanical pushbutton to DSO Channel 1 (unbounced); connect SR latch output to Channel 2; observe debouncing.

Diagnostic Troubleshooting of Digital Logic Circuits

- **Floating CMOS Inputs:** Leaving unused CMOS gates open causes intense power supply oscillations and random switching; tie all inputs.
- **Missing V_CC or Ground Pin:** IC may appear to partially function via parasitic input clamping diodes, but exhibits erratic, corrupted outputs.
- **Excessive Fan-Out Loading:** Connecting too many gate inputs to a single output exceeds I_OL / I_OH ratings; voltage levels degrade into invalid zone.
- **Propagation Delay Race Hazards:** Unequal signal propagation paths cause momentary false glitch pulses ('glitches'); add consensus gates.
- **Reversed IC Orientation:** Inserting a DIP-14 IC backwards (pin 1 where pin 14 belongs) connects V_CC to GND, destroying chip instantly.

Lecture 8 Self-Assessment & Upcoming Topics

SELF-ASSESSMENT & PROBLEM-SOLVING EXERCISES:

Q: Define an active-LOW logic signal and explain why open-collector / open-drain buses utilize active-LOW signaling.

Q: Prove algebraically and draw the schematic showing how a 2-input XOR gate is synthesized using exactly four 2-input NAND gates.

Q: Differentiate between a Half Adder and a Full Adder. Why is a Half Adder insufficient for adding multi-bit binary words?

Q: Explain the cause of the ripple-carry bottleneck in a parallel adder, and describe how Carry Lookahead logic solves it.

Q: Why is mechanical switch debouncing mandatory in agricultural electronic counters, and how does an SR latch eliminate bounce?

NEXT LECTURE PREVIEW & REQUIRED PREPARATION:

Lecture 9 will explore Boolean Algebra & Karnaugh Maps: Axioms, De Morgan's laws, SOP/POS, K-map minimization, and static hazards.

Lecture 8 Synthesis: Key Takeaways & Formula Sheet

CORE CONCEPTUAL TAKEAWAYS

✔️ Logic gates execute binary Boolean decisions based on well-defined voltage threshold levels (TTL vs CMOS).

✔️ NAND and NOR gates are functionally complete universal gates capable of synthesizing any logic function.

✔️ Propagation delay (t_pd) sets the fundamental upper limit on digital processing speed and clock frequencies.

✔️ Full Adders combine two Half Adders and an OR gate to add three bits including incoming carry C_in.

✔️ Bistable SR latches store one bit of digital memory and provide perfect bounce-free mechanical switch pulses.

CRITICAL ENGINEERING FORMULAS

★ AND Gate: $Y = A \cdot B$; OR Gate: $Y = A + B$

★ NAND Gate: $Y = /(A \cdot B)$; NOR Gate: $Y = /(A + B)$

★ XOR Gate: $Y = A \oplus B = /A \cdot B + A \cdot /B$

★ Universal NAND Inverter: $/A = /(A \cdot A)$

★ Universal NAND OR: $A + B = /(/A \cdot /B)$

★ Half Adder: $S = A \oplus B$; $C = A \cdot B$

★ Full Adder: $S = A \oplus B \oplus C_{in}$; $C_{out} = AB + C_{in}(A \oplus B)$

Prescribed References & Practical Laboratory Preparation

- **Mano, M. M., & Ciletti, M. D. (2018). Digital Design:** With an Introduction to Verilog HDL (6th ed.). Pearson.
- Floyd, T. L. (2015). Digital Fundamentals (11th ed.). Pearson.
- **Karim, M. A., & Chen, X. (2017). Digital Design:** Basic Concepts and Principles. CRC Press.

- **Laboratory Preparation:** Review DIP-14 pinout diagrams (74LS00, 74LS08, 74LS32, 74LS86) before Lab Session 8.
- **Field Assignment:** Complete Assignment 3, Questions 5 to 8 on logic gate synthesis and full adder circuits.